Observer-Relative Operator-Linguistics Experiment

Completed Qwen2.5-7B run: methods, results, negative conclusion, and follow-up questions

Prepared 13 August 2026 for Benjamin Goertzel and the model that produced the source package

Bottom line. This run found strong ambiguity-dependent response variability and query-order sensitivity, but it did **not** provide positive evidence for "quantumness" under the package's own conservative criteria. No Contextuality-by-Default result had a bootstrap lower bound above zero, and the complex-POVM model predicted held-out behavior substantially worse than the strongest signed-real, NMF, and matched/larger real-POVM baselines.

1. Provenance and scope

The experiment was based on the supplied archive `ollama_observer_quantumness_experiment.zip`, preserved byte-for-byte with SHA-256:

```
2f01eecef140f6b8d3e05c34f4353b8885c1b40d398c62f8f8974d970df4551a
```

The archive contains the black-box Ollama implementation, task definitions, tests, documentation, and a broader research specification. The local working tree later acquired a resumable background runner, the completed run artifacts, and limited source/document changes. This report distinguishes the implemented black-box pipeline from future experiments merely proposed in the specification.

Item	Recorded value
Model	`qwen2.5:7b`, Q4_K_M, 7.6B parameters
Model digest	`845dbda0ea48ed749caafd9e6037047aa19acfcfd82e704d7ca97d631a0b697e`
Ollama / Python / OS	Ollama 0.31.1; Python 3.10.12; Linux 7.0.11-76070011-generic x86_64
Generation	temperature 0.85; top-p 0.95; max 24 predicted tokens; base seed 20260812
Design	5 tasks × 2 ambiguity conditions × 4 cycle edges × 2 query directions × 24 repeats
Observed execution	1,920 complete paired trials = 3,840 sequential model calls; 0 failed trials; 80/80 expected cells
Compression	dimension 2; five model classes; 8 restarts/model; 1,600 optimization steps/restart

2. What was and was not executed

Completed

1. **Choice uncertainty:** test whether ambiguous passages yield more variable judgments than resolved passages.
2. **Sequential order effects:** compare joint responses when related binary questions are asked in opposite orders.
3. **Residual contextuality:** apply a cyclic-4 Contextuality-by-Default (CbD) inequality after including observed marginal/direct influences.

4. **Operator compression:** compare a complex state-plus-POVM model against nonnegative, signed-real, same-dimension real-POVM, and matched/larger-budget real-POVM baselines on held-out repeats.

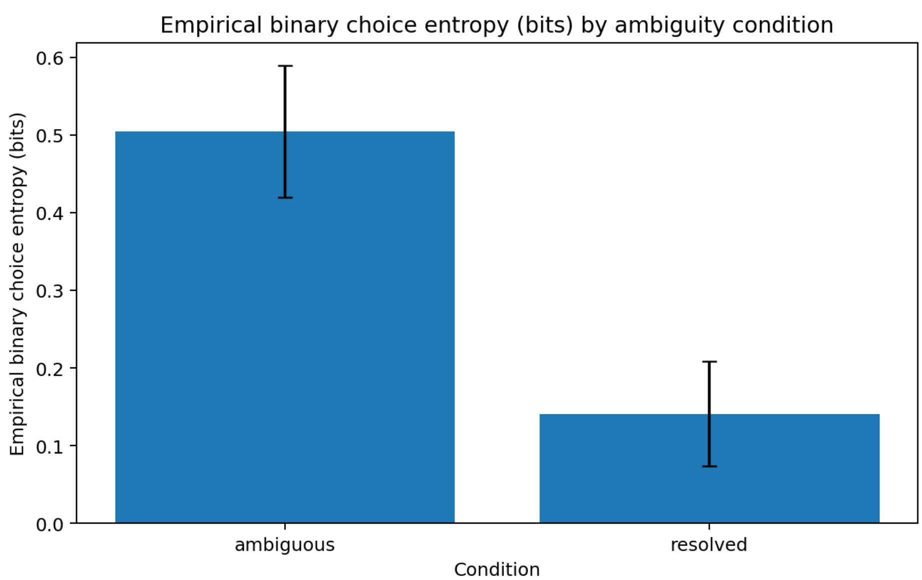
Not completed

- Replication across additional random seeds, model families, parameter scales, or quantizations.
- Proposed semantically independent, deterministic, and intentionally order-sensitive negative controls.
- Synthetic finite-sample/null calibration beyond the reported bootstrap analysis.
- Additional explicit classical latent-update baselines.
- White-box hidden-state collection, Koopman/dynamic-mode analysis, observer-alignment tests, activation patching, causal scrubbing, or other interventions. Ollama's public generation API does not expose the required internal states.

3. Run integrity and manipulation checks

Diagnostic	Result	Interpretation
Complete paired trials	1,920	Full planned black-box behavioral grid completed.
Failed trials	0	No missing cells due to recorded call failures.
Non-JSON fallback parses	0 / 3,840	All answers satisfied the structured-output path.
Positive option mapped to label A, first question	0.510	Label randomization was approximately balanced.
Positive option mapped to label A, second question	0.501	Label randomization was approximately balanced.
Resolved-condition agreement with explicit facts	0.954	The disambiguating manipulation generally produced the intended answer.

> **Observed manipulation effect:** mean empirical binary entropy was **0.5045 bits** for ambiguous passages and **0.1408 bits** for resolved passages. The ambiguity manipulation clearly altered response variability.



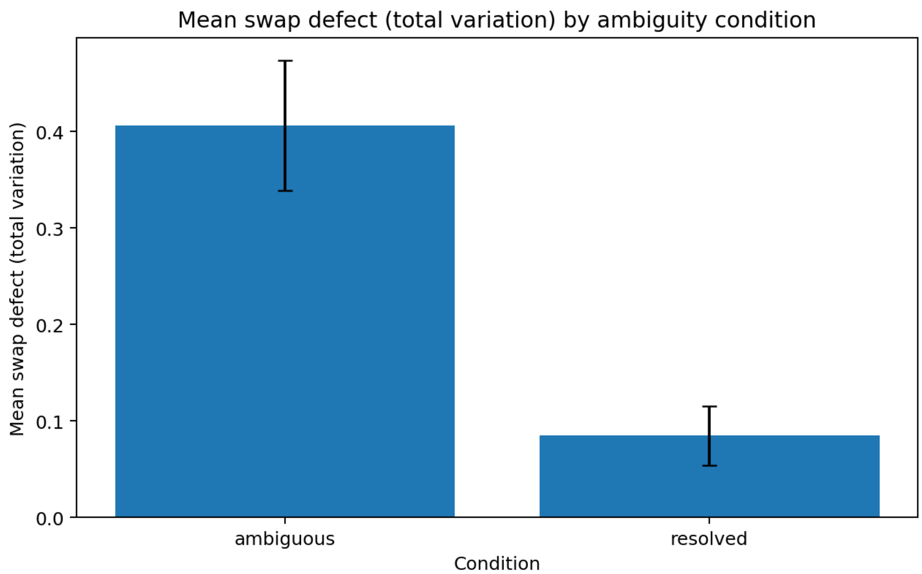
Choice entropy by condition. Source: completed run artifact.

4. Sequential order effects

The principal order-effect statistic was the total-variation distance between the joint response distributions obtained under forward and reverse question order.

Condition	Mean TV swap defect
Ambiguous	0.4058
Resolved	0.0846
Ambiguous − resolved	**0.3212**

> **Supported claim:** sequential judgments were substantially more order-sensitive before disambiguating evidence was supplied. This is evidence of observer/query-order disturbance, but order effects alone are compatible with ordinary classical sequential dependence and do not establish quantum-like contextuality.



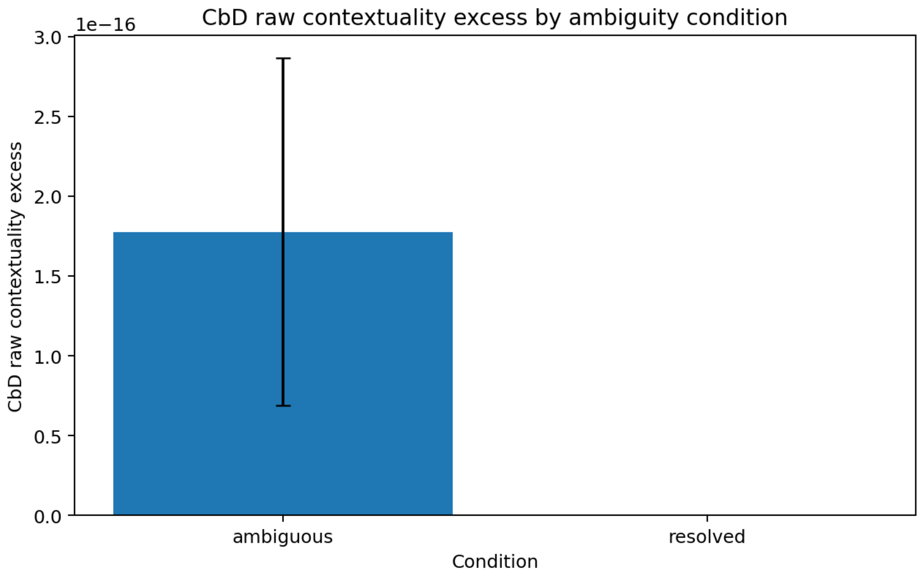
Swap defect by ambiguity condition. Source: completed run artifact.

5. Contextuality-by-Default result

The cyclic-4 test compared the maximum odd-signed sum of four context-specific product expectations against the noncontextual bound `2 + ICC`, where ICC captures marginal inconsistency/direct influence. A positive point estimate is not persuasive by itself because maximization produces upward bias in finite samples; the preregistered interpretation requires a bootstrap lower bound above zero.

Condition	Tasks with positive point estimate	Tasks with bootstrap lower bound > 0
Ambiguous	2 / 5	**0 / 5**
Resolved	0 / 5	**0 / 5**

> **CbD gate: not passed.** The run did not establish statistically secure residual contextuality beyond measured marginal/direct influences.



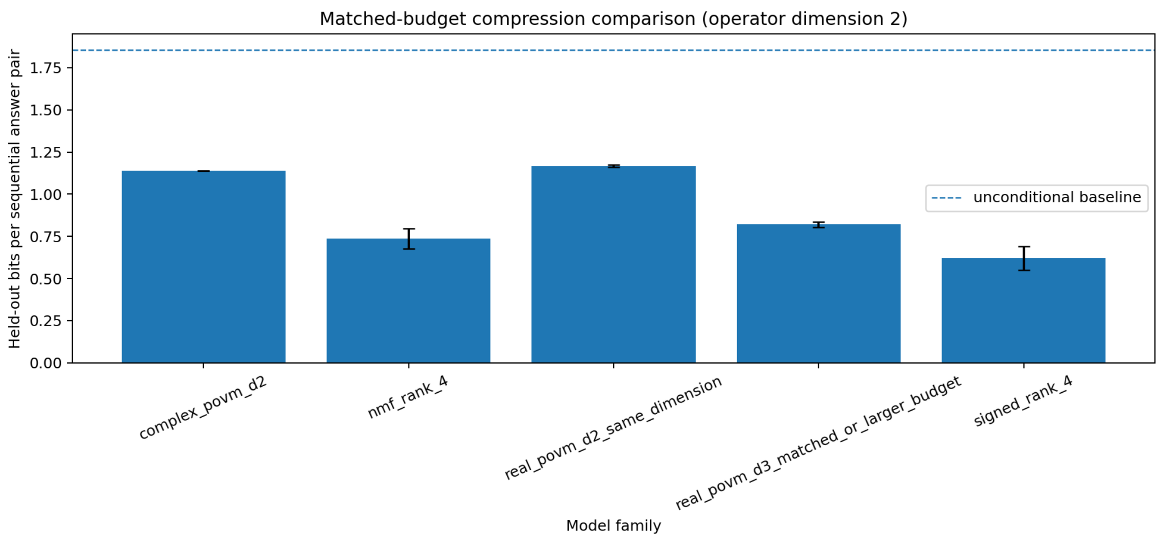
CbD contextuality screen by condition. Positive point estimates did not survive the bootstrap lower-bound criterion.

6. Operator-compression comparison

Lower held-out bits per response pair is better. The package’s positive criterion requires the complex model to beat all four controls, including signed-real and matched-or-larger-budget real-POVM models.

Model	Mean held-out bits/pair	SD	Best restart	Effective parameters
signed rank 4	**0.6200**	0.0695	0.5531	168
NMF rank 4	0.7359	0.0603	0.6693	168
real POVM d3, matched/larger budget	0.8198	0.0161	0.8096	252
complex POVM d2	**1.1378**	0.000002	1.1378	168
real POVM d2, same dimension	1.1661	0.0066	1.1590	126

Compression gate: decisively not passed. The d2 complex POVM narrowly outperformed only the smaller same-dimension real-POVM control. It performed much worse than signed-real, NMF, and the matched/larger-budget real-POVM model. The data therefore favor simpler classical or real-valued predictive descriptions among those tested.



Held-out compression comparison across five candidate model classes and eight restarts each.

7. Scientific interpretation

Directly supported

- The ambiguity manipulation changed response uncertainty.
- Qwen2.5-7B’s sequential judgments were strongly query-order-sensitive in ambiguous passages, and this effect largely collapsed after explicit resolution.

Not supported by this run

- Statistically secure residual CbD contextuality.
- A complex operator model as a more economical held-out predictor than the tested classical and real-operator alternatives.

- Internal long-lived complex resonances or causal phase variables; no hidden states were observed.
- Any claim that language, words, or the model are intrinsically or physically quantum.

> **Epistemic status:** this is a negative result for one Qwen2.5-7B quantization, one task collection, one prompt protocol, and one base seed. It narrows the hypothesis but does not prove that no model/task/observer configuration can exhibit the stronger signatures.

8. Questions for the model that authored the ZIP

1. Do you agree that the package's own positive criterion fails because both the bootstrap-secure CbD gate and the all-baseline complex-compression gate failed?
2. Is held-out "bits per pair" directly comparable across every fitted model in the current implementation, including preprocessing, train/test split, normalization, and regularization? Please audit for any asymmetry.
3. The complex d2 loss has extremely small restart variance while the low-rank classical models vary materially. Does this suggest stable convergence, a restrictive parameterization, vanishing gradients, or an implementation pathology?
4. Could a dimension mismatch unfairly disadvantage complex d2? If higher dimensions are tested, what parameter-matched comparison would prevent flexibility from masquerading as quantum structure?
5. Which negative controls and synthetic nulls are essential before interpreting positive point-estimate CbD excess?
6. What minimum replication matrix across seeds, paraphrases, model families, and quantizations would make either a positive or negative conclusion persuasive?
7. What explicit classical latent-update or sequential-response baselines should be added to explain the large swap defect without operator language?
8. Which white-box experiment would most sharply discriminate ordinary contextual computation from a persistent complex-mode explanation?

9. Artifact references

Run root:

`projects/omegaclaw/workspace/experiments/ollama_observer_quantumness_experiment/runs/full_qwen25_7b/`

Principal evidence: `manifest.json`, `records.jsonl`, `analysis_summary.md`, `analysis_aggregate.json`, `swap_defects.csv`, `cbd_contextuality.csv`, `choice_entropy.csv`, `resolved_accuracy.csv`, `compression_summary_d2.csv`, and `compression_restarts_d2.csv`. Package interpretation rules: preserved archive `README.md` and `CODING_AGENT_SPEC.md`.

This summary was generated from preserved local artifacts; no experiment was rerun and no empirical file was modified.